

Supplementary Material

Multi-Agent Forensic Reasoning for Generalizable Deepfake Video Detection

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A Additional Details of FaceVid-Forensics-100K

Table 1 summarizes the source- and method-level composition of the three splits. The real subset contains 21,075 videos: 12,934 from AIGVDBench (Ma et al. 2026), 7,058 from GenVidBench (Ni et al. 2026), 999 from FF++ (Rossler et al. 2019), and 84 from ViF-Bench (Li et al. 2026). After preprocessing and deduplication, 11,626 real videos are assigned to training, 3,733 to in-domain testing, and 5,716 to out-of-distribution (OOD) testing. For conventional face manipulations, videos are drawn from five manipulation methods in FF++ and grouped by manipulation type. The face-swapping (FS) subset includes Deepfakes, FaceSwap, and FaceShifter, with 1,288 videos assigned to training and 210 to in-domain testing. The face-reenactment (FR) subset includes Face2Face and NeuralTextures, with 858 videos assigned to training and 140 to in-domain testing.

The entire-face synthesis (EFS) subset is split at the generator level. Eight seen generators contribute 74,509 videos: Runway Gen-2 and Runway Gen-3 Alpha (Runway Research 2024), CausVid (Yin et al. 2025), CogVideo (Hong et al. 2023), ModelScope (Wang et al. 2023), Pika (Pika 2024), Text2Video-Zero (Khachatryan et al. 2023), and VideoCrafter2 (Chen et al. 2024a). Of these, 55,134 are for training and 19,375 for in-domain testing, with counts ranging from 456 (Runway Gen-2) to 32,690 (VideoCrafter2). The OOD split contains 1,920 fake videos from 20 additional generators drawn from the CogVideoX, Hunyuan-Video, LTX-Video, SkyReels, Wan, Runway, Hailuo, Kling, Pika, PixVerse, Sora, and Seedance families (Z.ai 2024; Kong et al. 2024; HaCohen et al. 2025; Chen et al. 2025; Wan Team et al. 2025; Runway Research 2025; MiniMax 2024; Kuaishou 2024; Pika 2025; PixVerse 2025; OpenAI 2025b; Team Seedance et al. 2026), ranging from 34 to 576 per generator. Spanning text-to-video, image-to-video, text-and-image-to-video, and video-to-video settings, these generators are excluded from training and in-domain testing. This generator-disjoint split evaluates generalization to unseen synthesis methods.

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B Additional Main Experimental Results

Detailed Out-of-Distribution Results

The per-generator accuracies in Table 2 indicate that the generalization benefit of the framework holds at the level of individual generators. After GRPO, our framework reaches a macro-average accuracy of 68.73% when the judge agent receives only the outputs of the four observation agents, and 69.47% when the sampled video frames are additionally provided. These two settings rank first and second on the macro average among all evaluated methods. With the sampled frames provided, our framework surpasses TALL++ (Xu et al. 2024), the strongest small vision model on average, by 5.25, and Skyra, the strongest forensics-tuned MLLM, by 4.24. Relative to the corresponding SFT settings, GRPO raises the macro average by 4.71 without video and 2.30 with video, whereas adding the sampled frames contributes 3.15 under SFT and 0.74 after GRPO. Both policy optimization and direct visual evidence therefore contribute to cross-generator generalization. With the sampled frames provided, our framework ranks first or second on 11 of the 20 held-out generators after GRPO, and the two GRPO settings together achieve the best result on six generators, which suggests that the improvement extends across several generator families rather than being driven by a single source.

In-Domain Results

Specialized small vision models remain the strongest detectors on seen distributions in Table 3, with TFCU (Guo et al. 2025) reaching 98.66% Acc and 99.46% F1. Among the MLLM-based methods, however, our framework performs best. After GRPO, it reaches 87.34% Acc and 96.35% F1 when the judge receives only the outputs of four observation agents, and 84.31% Acc and 96.19% F1 when the sampled video frames are additionally provided to the judge. Relative to the corresponding SFT settings, GRPO raises Acc by 9.15 and 5.23 and F1 by 1.05 and 0.80 in the two settings, respectively, while recall remains above 96.8%. Staged Observation–Judge training and policy optimization yield competitive performance on known manipulations, while our framework generalizes better to unseen generators and remains more interpretable than small vision baselines.

Category	Source	Train	In-Domain Test	Out-of-Distribution	Total
<i>Real Videos – 21,075</i>					
Real	AIGVDBench	9,706	3,228	0	12,934
	GenVidBench	1,061	365	5,632	7,058
	FF++	859	140	0	999
	ViF-Bench	0	0	84	84
<i>Fake Videos – 78,925</i>					
FS	Deepfakes	429	70	0	499
	FaceSwap	430	70	0	500
	FaceShifter	429	70	0	499
FR	Face2Face	429	70	0	499
	NeuralTextures	429	70	0	499
EFS	Runway Gen-2	320	136	0	456
	Runway Gen-3 Alpha	620	271	0	891
	CausVid	375	162	0	537
	CogVideo	1,787	595	0	2,382
	ModelScope	8,858	3,100	0	11,958
	Pika	11,232	3,931	0	15,163
	Text2Video-Zero	7,727	2,705	0	10,432
	VideoCrafter2	24,215	8,475	0	32,690
	CogVideoX1.5-5B-T	0	0	52	52
	HunyuanVideo	0	0	76	76
	HunyuanVideo-I2V	0	0	91	91
	LTX-Video-13B-I	0	0	80	80
	LTX-Video-13B-T	0	0	34	34
	SkyReels-V2	0	0	59	59
	SkyReels-V2-I2V-14B-540P	0	0	83	83
	Wan2.1-T2V-1.3B	0	0	76	76
	Wan2.1-VACE-1.3B-T	0	0	86	86
	Wan2.2-I2V-14B	0	0	86	86
	Wan2.2-T2V-14B	0	0	86	86
	Wan2.2-TI2V-5B-I	0	0	86	86
	Wan2.2-TI2V-5B-T	0	0	83	83
	Runway Gen-4 Turbo	0	0	49	49
	Hailuo AI	0	0	59	59
	Kling v1	0	0	62	62
	Pika 2.2	0	0	62	62
	PixVerse v4.5	0	0	64	64
	Sora 2	0	0	70	70
	Seedance 2.0	0	0	576	576
Total		68,906	23,458	7,636	100,000

Table 1: Composition of the proposed **FaceVid-Forensics-100K** dataset across the training, in-domain test, and out-of-distribution (OOD) splits, broken down by source and synthesis method. The fake subset contains face-swapping (FS), face-reenactment (FR), and entire-face synthesis (EFS) videos.

Qualitative Comparison

As shown in Figure 1, the fake video contains no obvious face-swapping boundary or severe frame-level distortion. GPT-4o (OpenAI 2024) and MiMo-V2.5 (Xiaomi MiMo Team 2026) therefore rely primarily on the stable facial appearance and the natural-looking outdoor illumination, interpreting the absence of conspicuous artifacts as evidence of

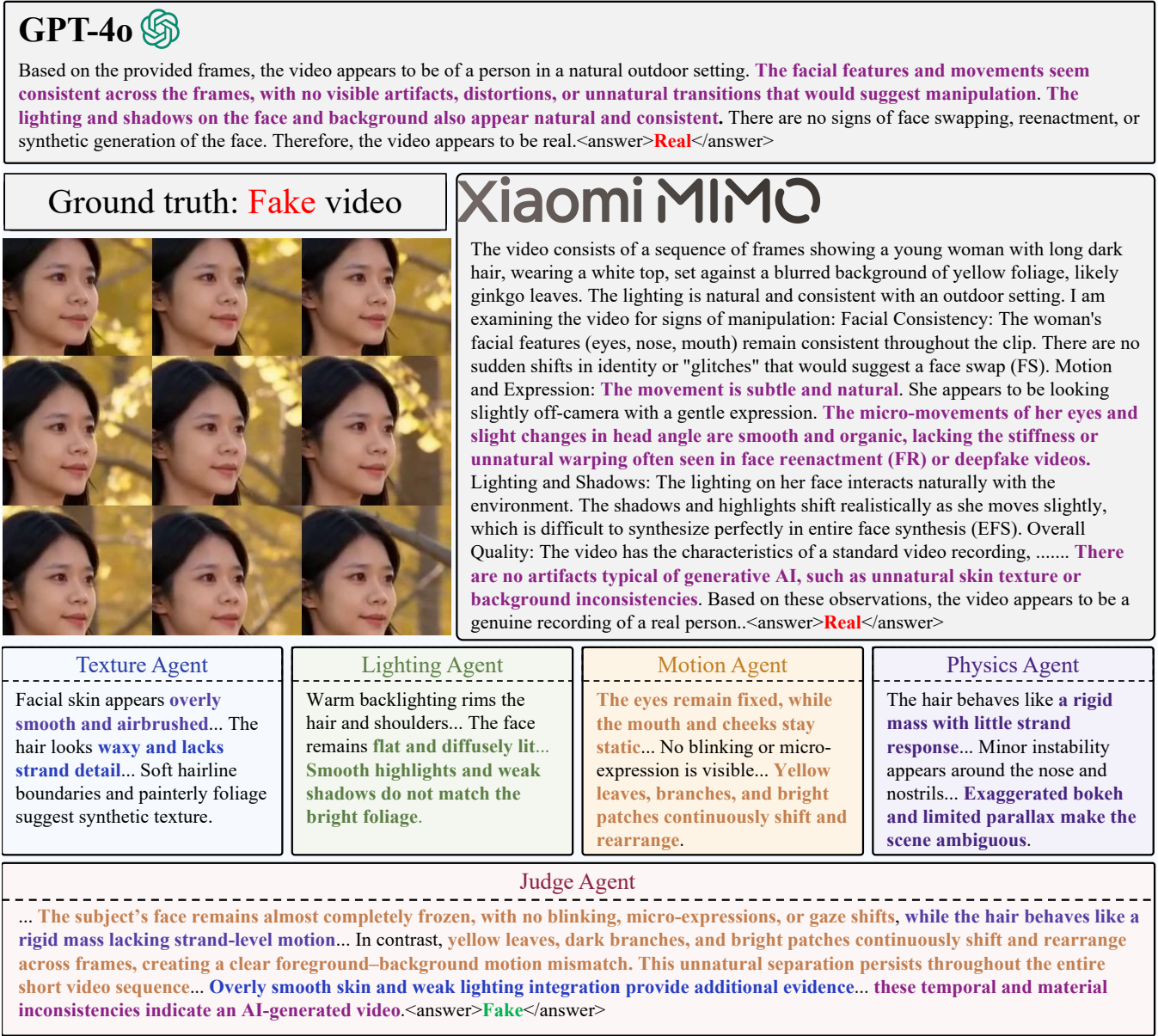
authenticity. Both models overlook the inconsistencies that become apparent only when the subject and the scene are compared across frames.

The four observation agents recover these weak but complementary cues. The texture observation agent identifies over-smoothed skin, waxy hair, soft hairline boundaries, and painterly foliage. The lighting observation agent notes that

Method	CogX	HunV	Hun2V	LTX-I	LTX-T	SkyV2	Sky2V	W21T	W21V	W22
<i>Small Vision Models</i>										
TALL++	79.49	70.93	47.48	60.74	71.40	<u>73.34</u>	52.23	68.30	62.92	50.71
TFCU	71.60	66.64	51.41	58.81	68.15	<u>62.81</u>	52.38	67.30	57.44	49.89
DFD-FCG	69.72	61.77	50.49	54.14	61.35	61.05	53.87	64.40	64.08	50.71
DFGaze	65.98	60.21	51.00	57.08	65.36	60.42	54.33	61.52	58.75	50.62
Effort	75.37	67.58	50.37	57.15	65.64	69.41	54.96	71.53	62.81	50.60
<i>Open-source MLLMs</i>										
Qwen3.5-0.8B	40.53	47.92	42.04	41.64	45.79	41.75	37.92	41.34	45.52	38.54
Qwen3.6-35B-A3B	51.22	53.80	49.57	46.89	50.20	44.24	48.02	51.17	48.49	46.16
MiMo-V2.5	51.01	48.28	50.60	47.98	45.30	45.74	47.17	46.96	46.42	47.59
<i>Closed-source MLLMs</i>										
GPT-5.5	47.96	48.72	46.31	45.99	44.11	47.50	47.13	46.74	47.60	46.44
Gemini 3.1 Pro	65.33	62.64	61.76	64.75	61.37	67.25	63.03	64.62	64.92	66.09
GPT-4o	58.60	57.18	58.33	59.32	53.62	57.79	54.92	59.82	57.21	58.96
Gemini 3.5 Flash	67.62	56.33	69.13	60.84	51.61	51.76	43.60	<u>70.81</u>	63.62	57.22
<i>Forensics-tuned MLLMs</i>										
Skyra	63.75	65.21	<u>66.91</u>	67.35	69.01	71.75	64.75	67.85	63.50	62.34
VideoVeritas	60.70	56.55	<u>53.01</u>	52.15	55.16	56.53	52.13	61.82	56.19	52.70
<i>Multi-Agent System</i>										
Ours (w/o Video, SFT)	67.68	67.23	57.65	61.86	68.08	63.17	64.06	64.59	66.68	56.79
Ours (w/o Video, +GRPO)	<u>79.14</u>	73.73	61.28	65.54	74.62	72.20	67.44	70.44	<u>70.96</u>	61.65
Ours (w/ Video, SFT)	69.30	69.20	62.71	<u>66.90</u>	67.26	68.06	64.11	67.22	69.66	60.94
Ours (w/ Video, +GRPO)	74.15	<u>73.69</u>	61.37	66.45	<u>71.60</u>	73.87	<u>65.70</u>	69.74	72.56	<u>62.67</u>

Method	W22T	W22I5	W22T5	Gen4	Hailuo	Kling	Plika2	PixV4.5	Sora2	Seed2.0
<i>Small Vision Models</i>										
TALL++	75.13	53.03	60.06	53.21	62.33	59.73	<u>81.50</u>	79.49	58.42	64.04
TFCU	59.19	51.05	65.63	56.74	56.03	62.88	83.04	70.22	49.70	73.77
DFD-FCG	57.11	59.43	68.93	50.72	61.05	57.12	77.29	57.58	50.21	66.09
DFGaze	62.24	50.62	53.73	57.91	56.18	59.80	75.13	60.21	49.85	54.13
Effort	66.88	56.41	73.03	53.67	63.48	56.20	77.98	66.84	53.67	67.79
<i>Open-source MLLMs</i>										
Qwen3.5-0.8B	40.29	36.22	45.15	42.67	36.67	41.62	44.04	37.89	38.69	39.97
Qwen3.6-35B-A3B	46.74	44.42	51.03	52.63	44.24	45.88	46.69	50.02	39.57	65.55
MiMo-V2.5	48.75	46.42	45.97	47.46	44.05	44.77	46.39	45.48	47.35	56.16
<i>Closed-source MLLMs</i>										
GPT-5.5	45.86	46.44	47.13	45.13	47.50	45.73	44.92	44.89	45.54	49.23
Gemini 3.1 Pro	62.02	63.76	67.25	61.13	62.17	65.64	63.22	64.91	62.96	66.64
GPT-4o	51.40	56.05	56.72	61.94	51.85	56.80	51.96	54.64	55.30	60.28
Gemini 3.5 Flash	61.87	68.27	<u>70.71</u>	54.37	59.39	63.80	70.26	60.53	61.11	67.30
<i>Forensics-tuned MLLMs</i>										
Skyra	66.99	60.01	61.14	69.97	68.36	72.09	59.19	66.57	69.76	48.17
VideoVeritas	56.77	48.63	57.55	50.03	50.60	54.69	57.91	62.15	52.68	64.49
<i>Multi-Agent System</i>										
Ours (w/o Video, SFT)	59.12	59.70	65.26	62.29	64.86	68.46	69.26	65.46	64.63	63.64
Ours (w/o Video, +GRPO)	62.24	65.14	67.44	60.42	<u>70.50</u>	<u>71.20</u>	76.85	70.85	67.77	65.29
Ours (w/ Video, SFT)	64.42	<u>66.75</u>	69.53	<u>67.02</u>	<u>67.21</u>	69.92	70.73	68.46	66.72	67.24
Ours (w/ Video, +GRPO)	<u>69.07</u>	61.51	70.52	61.61	73.87	69.28	74.92	<u>75.05</u>	<u>69.67</u>	<u>72.10</u>

Table 2: Detection accuracy on the 20 unseen video generators of the OOD test set, presented across two sub-tables (top: generators 1–10; bottom: generators 11–20). The best and second-best results are bolded and underlined, respectively.



Method	Acc	Recall	F1
Small Vision Models			
TALL++	96.06	<u>99.36</u>	99.00
TFCU	98.66	99.31	99.46
DFD-FCG	97.86	99.07	99.22
DFGaze	<u>98.52</u>	<u>98.51</u>	<u>99.11</u>
Effort	<u>97.75</u>	99.50	<u>99.37</u>
Open-source MLLMs			
Qwen3.5-0.8B	53.99	55.39	67.40
Qwen3.6-35B-A3B	74.84	83.33	87.85
MiMo-V2.5	70.52	51.32	66.97
Closed-source MLLMs			
GPT-5.5	65.05	38.16	54.64
Gemini 3.1 Pro	61.57	94.58	90.90
Forensics-tuned MLLMs			
Skyra	48.26	3.83	7.29
VideoVeritas	61.31	88.20	87.93
Multi-Agent System			
Ours (w/o Video, SFT)	78.19	98.22	95.30
Ours (w/o Video, +GRPO)	87.34	96.87	96.35
Ours (w/ Video, SFT)	79.08	98.07	95.39
Ours (w/ Video, +GRPO)	84.31	97.78	96.19

Table 3: Overall performance on the in-domain test set. The best and second-best results are highlighted in bold and underlined.

set, obtained by randomly sampling 100 real and 100 fake videos. For each detector output, two independent judges—the open-source DeepSeek-V4 Pro (DeepSeek-AI et al. 2026) and the closed-source GPT-5-mini (OpenAI 2025a)—score the generated rationale, so that the assessment does not rest on a single model family or provider. Both judges receive the same ground-truth answer, reference rationale, model answer, and model rationale, and both apply an identical rubric covering evidence accuracy, reference alignment, specificity and grounding, clarity and conciseness, and consistency between the rationale and the verdict. Each judge returns a single holistic integer score from 1 to 10. Table 4 reports the mean scores for real videos, fake videos, and the complete 200-video subset. The shared evaluator prompt is provided in Listing 8.

DeepSeek-V4 Pro assigns the highest overall score (6.37) to our SFT configuration with video input, while GPT-5-mini assigns it 6.25. Within our framework, providing video input improves the overall score both with and without GRPO optimization across both judges, suggesting better grounding of generated rationales in visual evidence. GRPO-optimized configurations remain competitive, although GRPO primarily targets classification performance rather than the explanation-quality rubric; therefore, these scores should be interpreted together with detection performance. All four configurations receive higher scores on fake than on real videos, but this pattern is specific to the sampled subset and should not be generalized more broadly.

C Reasoning Strategies

Figure 2 summarizes the four reasoning structures evaluated in our experiments. Single directly predicts the video label from the sampled frames. CoT uses a single agent and a single interaction, but explicitly prompts the model to reason step by step over the four forensic dimensions before giving the final verdict. Multi-turn dialogue instead queries the same agent through multiple turns: dimension-specific observations are collected in earlier turns and then synthesized into a final prediction in a subsequent turn. In contrast, our multi-agent system assigns the four forensic dimensions to independent observation agents and uses a separate judge agent to reconcile their evidence and issue a final verdict.

Single. The Single baseline maps the sampled frames directly to an explanation and a prediction in a single model call. The general-purpose open- and closed-source MLLMs use the shared prompts in Listing 1, with the sampled frames and their timestamps appended to the user message.

Chain-of-Thought. The CoT baseline performs the four forensic analyses and produces the final decision in a single model call. Its exact system and user prompts are shown in Listing 2.

Multi-turn. The Multi-turn baselines use a shared system prompt and five user turns: the first four turns examine texture, lighting, motion, and physical plausibility, respectively, while the fifth turn produces the final explanation and verdict. The three variants differ only in when video frames are provided to the model, as summarized in Table 5. In *Multi-turn*, frames are provided only in the first turn, and all subsequent turns rely on the retained dialogue context and text-only prompts. In *Multi-turn-Obs*, frames are provided during the four observation turns, allowing each aspect-specific analysis to directly access the visual evidence, whereas the final verdict turn is text-only and aggregates the preceding observations. In *Multi-turn-All*, frames are provided in all five turns, including the final verdict turn, so the model can revisit the visual evidence when generating the explanation and prediction. All three settings use the same first four prompts in Listing 3 and differ only in the final prompt, as shown in Listing 4.

Setting	Turns with Frames	Turn-5 Prompt
Multi-turn	1	Text-only
Multi-turn-Obs	1–4	Text-only
Multi-turn-All	1–5	Video-aware

Table 5: Frame-delivery settings for Multi-turn inference. Turns not listed in the middle column receive text only, and the preceding textual responses remain in the dialogue history in every setting.

Efficiency Analysis

We evaluate latency and throughput on 100 randomly sampled OOD videos (50 real and 50 fake) using a single NVIDIA RTX 5090 GPU. As shown in Table 6, Single and CoT have the lowest latency, while Multi-turn has the highest latency on both MLLMs. The latency of our two

Method	DeepSeek-V4 Pro			GPT-5-mini		
	Fake	Real	Overall	Fake	Real	Overall
Open-source MLLMs						
Qwen3.6-35B-A3B	3.49	5.30	4.39	5.02	6.19	5.61
MiMo-V2.5	2.00	6.84	4.42	3.76	<u>7.30</u>	5.53
Closed-source MLLMs						
GPT-4o	2.65	4.56	3.60	4.37	6.07	5.22
Gemini 3.1 Pro	7.15	3.65	5.40	<u>7.45</u>	5.02	6.24
Gemini 3.5 Flash	5.98	6.52	<u>6.25</u>	6.64	7.12	6.88
Forensics-tuned MLLMs						
Skyra	3.16	<u>6.58</u>	4.87	4.70	7.35	6.03
VideoVeritas	8.05	<u>4.07</u>	6.06	7.90	5.31	<u>6.61</u>
Multi-Agent System						
Ours (w/o Video, SFT)	7.66	4.58	6.12	6.99	5.12	6.05
Ours (w/o Video, SFT+GRPO)	6.40	4.46	5.43	6.67	4.85	5.76
Ours (w/ Video, SFT)	<u>7.89</u>	4.84	6.37	7.27	5.22	6.25
Ours (w/ Video, SFT+GRPO)	7.75	4.14	5.95	7.37	4.68	6.03

Table 4: Mean explanation-quality scores on a subset of the OOD test set containing 100 randomly selected real and 100 randomly selected fake videos. The best and second-best results in each column are bolded and underlined.

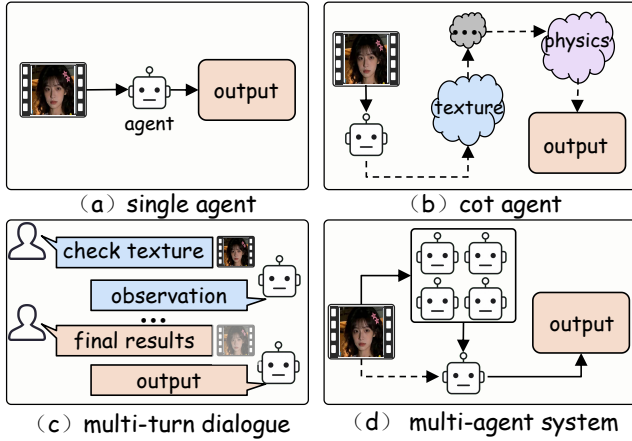


Figure 2: Comparison of the reasoning strategies evaluated in our experiments: (a) single-agent direct prediction, (b) single-agent chain-of-thought reasoning, (c) multi-turn dialogue, and (d) our multi-agent system with four independent observation agents and a separate judge agent.

configurations lies between these baselines, and both are faster than Multi-turn. This difference is particularly clear on InternVL3.5-8B, where our latency is 4.781–5.486 seconds per video, compared with 14.150 seconds for Multi-turn. Our framework also achieves 151.293–181.88 tokens per second across the two MLLMs, approximately twice the throughput of Multi-turn. Thus, although our framework is slower than the one-pass strategies, parallel execution of the observation agents keeps its additional inference cost moderate. The configuration without video input further reduces latency by avoiding video processing in the judge agent.

Strategy	Latency ↓ (s/video)	Throughput ↑ (tokens/s)
Qwen2.5-VL-7B		
Single	1.503	79.10
CoT	2.095	85.27
Multi-turn	7.123	90.47
Ours (w/ Video)	6.128	172.26
Ours (w/o Video)	5.791	181.88
InternVL3.5-8B		
Single	1.583	53.892
CoT	2.864	70.869
Multi-turn	14.150	84.581
Ours (w/ Video)	5.486	151.293
Ours (w/o Video)	4.781	168.329

Table 6: Efficiency comparison of different reasoning strategies on an NVIDIA RTX 5090 GPU.

Qualitative Comparison of Reasoning Strategy

As shown in Figure 3, Single prediction, CoT, and Multi-turn dialogue all interpret the smooth facial appearance, stable ge-

Single (Qwen2.5-VL-7B) ...hand gesture consistent across frames...no motion blur or inconsistencies...lighting consistent...no obvious artifacts...frames appear real but evidence limited...<answer>Real</answer>	Chain of Thought (CoT) ...smooth skin texture...no artifacts or inconsistencies...consistent light source direction...no temporal flickering...natural hand gestures...hair dynamics consistent...occlusion ordering is correct...facial geometry stable...texturally and physically consistent...<answer>Real</answer>		
Ground truth: Fake video 	Multi-turn dialogue Turn 1 👤: ...From the perspective of texture, observe the video... 🤖: ...skin texture looks natural...no significant blending artifacts... Turn 2 👤: ...From the perspective of lighting, observe the video... 🤖: ...Highlight consistency is maintained...soft shadows falling naturally... Turn 3 👤: ...From the perspective of motion, observe the video... 🤖: ...inter-frame motion continuity...hand gesture remains stable... Turn 4 👤: ...From the perspective of physics, observe the video... 🤖: ...hair dynamics appear natural...Facial geometry looks consistent... Turn 5 👤: ...Based on your four observations above... 🤖: ...all coherent and realistic...<answer>Real</answer>		
Texture Agent ...an unnaturally smooth, waxy quality with a lack of fine pores and micro-textures...severe blending artifacts...unnatural warping...overly bright and lacking natural shading...	Lighting Agent ...the facial lighting exhibits an unnaturally uniform quality...The highlights...appear overly smooth and flat...the specular highlights on the teeth and lips are notably bright and uniform...	Motion Agent ...fingers appear to morph and blend together rather than maintaining distinct anatomical structure...with strands warping and re-forming between frames...minor micro-jitters and shape-shifting...	Physics Agent ...the hair remains largely static and lacks natural secondary motion...fingers appearing to merge, stretch, and lose structural integrity...proper depth separation...
Judge Agent ...the complete absence of biological motion in the subject's eyes...they remain wide open and fixed in a glassy stare throughout the clip, lacking the involuntary saccades, blinks, or micro-movements inherent to a living person. This frozen gaze...The facial expression itself is unnaturally static...The skin texture is overly smooth and waxy, devoid of fine pores and imperfections...the hand gesture...lacks the subtle, dynamic tension of a real human hand, appearing more like a digital overlay...these are superficial cues...outweighed by the profound failures in simulating basic human physiology and anatomy...<answer>Fake</answer>			

Figure 3: Qualitative comparison of Single prediction, CoT, Multi-turn dialogue, and our multi-agent system (MAS) on a fake video. Single, CoT, and Multi-turn cite plausible but superficial evidence of authenticity and incorrectly predict Real. In contrast, the independent observation agents identify complementary anomalies in texture, lighting, motion, and physical plausibility, which the judge agent reconciles to correctly predict Fake.

ometry, coherent illumination, and apparently natural motion as evidence that the video is real. Specifically, Single prediction relies on a coarse overview and prematurely declares the video authentic despite acknowledging limited evidence. Step-by-step reasoning via CoT or decomposing the analysis into successive dialogue turns fails to correct this initial assessment: the single-agent trajectory repeatedly reinforces benign interpretations across all forensic dimensions, ultimately repeating the incorrect Real verdict.

Our MAS instead elicits complementary evidence from

independent observation agents. The texture agent identifies waxy skin, missing micro-texture, and blending artifacts; the lighting agent reports unnaturally uniform facial illumination; the motion agent detects morphing fingers and temporal shape-shifting; and the physics agent highlights static hair and implausible hand dynamics. The judge agent reconciles these observations and further emphasizes the frozen gaze and the absence of natural biological micro-motion, which yields the correct Fake prediction. This example illustrates that the advantage of the MAS comes not merely from per-

forming more reasoning steps, but from collecting diverse forensic evidence independently and reconciling it explicitly before prediction.

D Observation and Judge Model Combinations

Observation	Judge	Acc	Recall	F1
w/o Video				
InternVL3.5	InternVL3.5	62.72	<u>85.52</u>	46.94
InternVL3.5	Qwen2.5-VL	<u>63.50</u>	85.15	47.54
Qwen2.5-VL	InternVL3.5	60.70	86.93	45.51
Qwen2.5-VL	Qwen2.5-VL	63.61	82.24	<u>47.53</u>
w/ Video				
InternVL3.5	InternVL3.5	69.07	<u>83.80</u>	52.43
InternVL3.5	Qwen2.5-VL	67.95	78.33	51.43
Qwen2.5-VL	InternVL3.5	<u>68.89</u>	82.45	<u>52.28</u>
Qwen2.5-VL	Qwen2.5-VL	67.03	87.76	50.49

Table 7: Performance of the Observation–Judge model combinations on the OOD test set. InternVL3.5 is InternVL3.5-8B and Qwen2.5-VL is Qwen2.5-VL-7B. All configurations use SFT only.

We further evaluate whether the observation agents and the judge agent need to come from the same MLLM family. As shown in Table 7, heterogeneous combinations remain competitive with homogeneous ones. When the judge agent receives only the observation agent outputs, InternVL3.5 (Chen et al. 2024b) observations judged by Qwen2.5-VL (Bai et al. 2025) achieve the best F1 of 47.54%. When the sampled video frames are additionally provided, InternVL3.5 used for both components performs best, reaching 69.07% Acc and 52.43% F1, while the heterogeneous Qwen2.5-VL–InternVL3.5 combination achieves comparable results. Providing the sampled frames to the judge agent improves Acc and F1 for every combination, indicating that the framework is modular with respect to the MLLM used for each component.

E Impact of Model Parameter Scale

We evaluate the scalability of our framework across different parameter scales using the 3B, 7B, and 32B variants of Qwen2.5-VL. To isolate the effect of model capacity, all variants are trained under the SFT protocol across both Judge Agent settings, in which the Judge Agent receives either only the observation reports or these reports together with the sampled video frames.

As shown in Table 8, scaling the base MLLM from 3B to 32B parameters yields consistent gains in accuracy and F1 score. When the Judge Agent receives only the observation reports, accuracy increases from 61.36% (3B) to 63.61% (7B) and reaches 66.40% (32B), while F1 score steadily improves from 45.81% to 49.92%. A similar scaling trend is observed when the sampled video frames are additionally provided, with F1 score reaching a peak of 50.81% at 32B.

Notably, as model capacity grows, the text-only Judge Agent becomes increasingly adept at reconciling textual forensic reports, narrowing the gap with the video-aware setting while maintaining solid interpretability.

Parameter	w/o Video			w/ Video		
	Acc	Recall	F1	Acc	Recall	F1
3B	61.36	83.33	45.81	66.65	<u>82.55</u>	50.16
7B	<u>63.61</u>	82.24	<u>47.53</u>	<u>67.03</u>	87.76	<u>50.49</u>
32B	66.40	80.36	49.92	67.36	81.47	50.81

Table 8: Performance comparison across different Qwen2.5-VL parameter scales on the OOD test set under SFT. The best and second-best results within each Judge Agent setting are highlighted in bold and underlined.

F Prompt Templates

Agent	Prompt-specific fields
Texture	expert/perspective: texture and detail; focus: texture; cues: skin texture, edge sharpness, blending artifacts, material consistency, and fine-grained detail stability across frames
Lighting	expert/perspective: lighting; focus: lighting; cues: light source direction, highlight consistency, shadow placement, specular reflections, and overall illumination coherence
Motion	expert/perspective: motion; focus: motion; cues: inter-frame motion continuity, unnatural movements, temporal flickering, or physically implausible actions
Physics	expert/perspective: physical plausibility; focus: physics; cues: hair dynamics, clothing behavior, occlusion ordering, perspective correctness, and geometric deformation

Table 9: Role-specific fields used to instantiate the shared Observation Agent prompt templates of Listing 5 for each of the four forensic dimensions.

Multi-Agent System. Each observation agent receives the sampled video frames and a prompt tailored to one forensic dimension. The four prompts share the templates in Listing 5, and Table 9 provides the exact role-specific fields. The judge agent then receives the four observations through a shared user template. Without video input, the judge agent receives only these observations; with video input, it additionally receives the sampled frames. The corresponding system prompts are shown in Listing 6.

Observation Label Generation. For textual label construction, the annotator MLLMs use the same dimension-specific prompts given in Listing 5 and Table 9. Their user messages additionally contain the uniformly sampled frames and the corresponding timestamps. For each dimension, DeepSeek-V4 Pro then combines the available model reports using the prompts in Listing 7. The ground-truth label is provided only as guidance for resolving conflicting reports, and the output is required to contain neither a verdict nor a reference to that label.

Explanation Evaluation. DeepSeek-V4 Pro and GPT-5-mini independently evaluate the generated rationales using the same inputs, rubric, and output format. Each judge receives the ground-truth answer, a reference rationale, the model answer, and the model rationale, and assigns a holistic integer score from 1 to 10 based on evidence accuracy, reference alignment, specificity and grounding, clarity and conciseness, and verdict consistency. The shared evaluator prompt is shown in Listing 8, and the sampling protocol and results are reported in Section B.

Listing 1: Prompts used by the Single baseline for the general-purpose open- and closed-source MLLMs.

System Prompt

You are an expert video analyst. Please think about the question as if you were a human pondering deeply. It’s encouraged to include self-reflection or verification in the reasoning process. Put the explanation of your judgment within <explanation></explanation> tags. Finally, give the final verdict within <answer></answer> tags.

User Prompt

Is this video real or fake?

The following images are uniformly sampled frames from the video.

Listing 2: Prompts used by the chain-of-thought (CoT) baseline.

System Prompt

You are an expert face video forensics analyst. You are shown frames uniformly sampled from a video, in order. Determine whether the face video is Real or Fake based ONLY on the visible evidence in these frames.

Think step by step. Work through the following four analysis steps in order. In each step, report concrete, specific visual cues - mention frame ranges, facial regions, or objects where you see potential manipulation artifacts, or where everything appears consistent and natural. Do NOT give a real/fake verdict inside these four steps.

1. Texture and detail: skin texture, edge sharpness, blending artifacts around the face boundary, material consistency, and fine-grained detail stability across frames.
2. Lighting: light source direction, highlight consistency, shadow placement, specular reflections on skin and eyes, and overall illumination coherence between the face and the scene.
3. Motion: inter-frame motion continuity, unnatural movement, temporal flickering or jitter, and physically implausible actions across frames.
4. Physical plausibility: hair dynamics, clothing behavior, occlusion ordering, perspective correctness, facial geometry, and geometric deformation.

After the four steps, weigh the four observations together and reach a final decision. If clear manipulation artifacts appear in one or more steps, answer fake. If the face is texturally, temporally, and physically consistent with the scene, answer real.

System Prompt (continued)

Output your response strictly in the following format, with every tag present and non-empty:

<texture>your texture and detail observations</texture>
<lighting>your lighting observations</lighting>
<motion>your motion observations</motion>
<physics>your physical plausibility observations</physics>
<explanation>a brief rationale that synthesizes the four observations above into your decision</explanation>
<answer>real</answer>

Output requirements:

- * All six tags are required and must not be empty.

- * Each of <texture>, <lighting>, <motion>, <physics> must contain at least one concrete visual observation and must NOT contain a real/fake judgment.
- * <explanation> must contain at least one sentence and must be grounded in the observations above.
- * The answer must be exactly either <answer>real</answer> or <answer>fake</answer>, with no other text inside the answer tag.
- * Base every observation only on the visible evidence in the frames. Do not use audio, speech, or metadata as evidence.
- * Do not output anything outside these tags.

User Prompt

Is this face video real or fake?

The following images are uniformly sampled frames from the video, in order.

Listing 3: System prompt and four observation-turn prompts shared by all three Multi-turn frame-delivery settings.

System Prompt

You are an expert face video forensics analyst. Your task is to determine whether the face video is real or fake based only on the visible evidence in the provided frames.

Turn 1: Texture and Detail

These are frames uniformly sampled from the video, in order. From the perspective of texture and detail, observe the video and report any anomalies related to skin texture, edge sharpness, blending artifacts around the face boundary, material consistency, and fine-grained detail stability across frames. Be concrete - mention frame ranges, facial regions, or objects. Put your report inside <observation></observation> tags. Do not make a real/fake judgment yet.

Turn 2: Lighting

Now from the perspective of lighting, report any anomalies related to light source direction, highlight consistency, shadow placement, specular reflections on skin and eyes, and overall illumination coherence between the face and the scene. Be concrete - mention frame ranges, facial regions, or objects. Put your report inside <observation></observation> tags. Do not make a real/fake judgment yet.

Turn 3: Motion

Now from the perspective of motion, report any anomalies related to inter-frame motion continuity, unnatural movements, temporal flickering or jitter, and physically implausible actions across frames. Be concrete - mention frame ranges, facial regions, or objects. Put your report inside <observation></observation> tags. Do not make a real/fake judgment yet.

Turn 4: Physical Plausibility

Now from the perspective of physical plausibility, report any anomalies related to hair dynamics, clothing behavior, occlusion ordering, perspective correctness, facial geometry, and geometric deformation. Be concrete - mention frame ranges, facial regions, or objects. Put your report inside <observation></observation> tags. Do not make a real/fake judgment yet.

Listing 4: Final-turn prompts for the three Multi-turn frame-delivery settings.

Text-only Verdict (Multi-turn, Multi-turn-Obs)

Based on your four observations above, decide whether the face video is real or fake. First put a brief explanation that synthesizes your four observations within <explanation></explanation> tags; do not put any answer or label inside the explanation. Then output exactly either <answer>real</answer> or <answer>fake</answer>, with no other text inside the answer tag.

Image-aware Verdict (Multi-turn-All)

Based on your four observations above and the video frames, decide whether the face video is real or fake. First put a brief explanation that synthesizes your four

```
observations within <explanation></explanation> tags;
do not put any answer or label inside the explanation.
Then output exactly either <answer>real</answer> or <
answer>fake</answer>, with no other text inside the
answer tag.
```

Listing 5: Prompt templates shared by the four observation agents. The fields in braces are instantiated with the role-specific values listed in Table 9.

Observation agent system prompt

```
You are a {expert} analysis expert in a video forensics
team. Your task is to carefully observe the video and
report any anomalies related to {cues}. Focus ONLY on {
focus}-related observations. Do NOT make a final real/
fake judgment. Return only a concise report inside <
observation></observation> tags.
```

Observation agent user prompt

```
Observe the provided video from the perspective of {
perspective} analysis. List the specific visual cues you
noticed. Be concrete and precise - mention frame ranges
, regions, or objects where you see potential issues or
where everything appears normal. Put your report inside
<observation></observation> tags. Do not make a final
real/fake judgment and do not use <answer> tags. Keep
your response concise (within 200 words).
```

Listing 6: Judge agent prompts with and without video input. Both settings use the same user prompt, populated with the four observation agent reports; the sampled frames are additionally attached when video input is provided.

Judge System Prompt: w/o Video

```
You are the final judge for binary video forgery
classification. You cannot access any images or video.
Use only the four expert analysis reports supplied as
text. Put a brief explanation of your judgment within <
explanation></explanation> tags; do not put any <answer>
tags or final label inside the explanation. Then output
exactly one tagged lowercase label: <answer>real</
answer> or <answer>fake</answer>.
```

Judge System Prompt: w/ Video

```
You are the final judge for binary video forgery
classification. You are given the video frames together
with four expert analysis reports (texture, lighting,
motion, physical plausibility) supplied as text. Weigh
both the visual evidence and the four reports. Put a
brief explanation of your judgment within <explanation
></explanation> tags; do not put any <answer> tags or
final label inside the explanation. Then output exactly
one tagged lowercase label: <answer>real</answer> or <
answer>fake</answer>.
```

Shared Judge User Prompt

```
Texture analysis report:
{texture_report}

Lighting analysis report:
{lighting_report}

Motion analysis report:
{motion_report}

Physical plausibility analysis report:
{physics_report}

Return a brief explanation in <explanation></explanation
> tags. Do not put any <answer> tags or final label
inside the explanation. Then output exactly <answer>real
</answer> or <answer>fake</answer>.
```

Listing 7: DeepSeek-V4 Pro prompts for aggregating the annotator reports of one forensic dimension into a single observation label. The aggregation is performed independently for texture, lighting, motion, and physics.

DeepSeek-V4 Pro System Prompt

```
You are a video forensics expert. Multiple AI models
have each analyzed the same video from a specific
analytical perspective and produced independent
```

```
observation reports. Your task is to read all of their
reports and synthesize them into a single, coherent,
comprehensive observation.
```

```
Aggregate multi-agent observations with the video label
(real or fake) in mind: use the ground-truth label
solely as internal guidance to resolve conflicts between
models when their reports disagree.
```

Rules:

- Write ONE unified observation paragraph (100-200 words).
- Do NOT list models by name (e.g. "Model 1 said..."). Present the observations as a single cohesive expert analysis.
- Resolve conflicts: when models disagree, prioritise observations consistent with the video label and note discrepancies as ambiguous cues.
- Focus on concrete forensic signals and visual evidence mentioned by the models.

DeepSeek-V4 Pro System Prompt (continued)

- Do NOT output a real/fake verdict. Only output the observation text.
- Do NOT mention or reference the ground-truth label, the video's real/fake status, or whether the video is real or fake in your output. The observation must read as if the observer does not know the verdict.
- Do NOT wrap your output in any tags. Output plain text only.

DeepSeek-V4 Pro User Prompt Template

```
This video is {ground_truth}.
Below are {n} observation reports from forensic models
that analyzed the {dimension_name} of this video. Each
report focuses on {dimension_description}.
```

```
--- Model 1 ({model_name_1}) ---
{observation_1}

...

--- Model {n} ({model_name_n}) ---
{observation_n}
```

Listing 8: Prompt shared by DeepSeek-V4 Pro and GPT-5-mini for evaluating the generated forensic explanations, following the explanation evaluation prompt of VidGuard-R1. Each judge returns a concise analysis and a holistic integer score from 1 to 10.

Evaluator System Prompt

```
You are an independent expert evaluating the quality of
an explanation
generated by a face-video forgery detection model. The
model predicts whether
a video is real or fake and provides a rationale for its
prediction.
```

```
Evaluate the rationale with respect to forensic evidence
involving texture and
detail, lighting, motion and temporal consistency, and
physical plausibility.
```

Consider the following five criteria:

1. Evidence accuracy:
Does the rationale identify relevant signs of manipulation or authenticity?
2. Reference alignment:
Does it agree with the important visual evidence described in the reference rationale?
3. Specificity and grounding:
Does it provide concrete, video-related observations rather than vague claims, unsupported assumptions, or hallucinated details?

Evaluator System Prompt (continued)

4. Clarity and conciseness:
Is the rationale coherent, direct, and free from unnecessary repetition?

```

5. Verdict consistency:
Does the rationale logically support the model's
predicted real/fake label?
A rationale that contradicts the ground-truth label or
its own predicted label
should not receive a high score.

Assign one holistic integer score from 1 to 10:
- 1 indicates an incorrect, unsupported, or
contradictory rationale.
- 10 indicates an accurate, specific, concise rationale
that fully aligns with
the reference evidence and convincingly supports the
correct verdict.

Return exactly two lines:

ANALYSIS: <a concise paragraph evaluating the rationale>
SCORE: <an integer from 1 to 10>

Do not output anything else.

```

Evaluator User Prompt Template

```

Evaluate the following model-generated explanation.

Ground-truth answer:
{ground_truth_answer}

Reference rationale:
{ground_truth_rationale}

Model answer:
{model_answer}

Model rationale:
{model_rationale}

```

candidates increasingly agree, and it falls to nearly zero with video input over the final third of training but remains higher without video.

G Training Dynamics

Figure 4(a–d) shows the SFT dynamics. All four observation agents converge in the same manner: the training loss drops steeply within roughly the first 10% of training and then declines gradually, while token accuracy rises to a plateau that is essentially flat over the second half of training. The four agents remain separated by a small and stable margin, with the lighting observation agent attaining the lowest loss and the highest token accuracy and the physics observation agent the highest loss and the lowest token accuracy. The two judge agent configurations converge to nearly the same point, with training losses approaching 1.2 and token accuracies approaching 0.66. They differ mainly in how quickly they get there: with video input the loss falls faster over the first 20% of training, and its curve remains slightly more oscillatory thereafter, whereas the two configurations are largely indistinguishable once training passes the halfway point.

Figure 4(e–g) shows the GRPO dynamics of the judge agent. The mean reward rises sharply within early training and fluctuates around a high plateau, with the two configurations closely aligned and neither consistently ahead; because the reward is the binary accuracy of the parsed verdict, this indicates that GRPO refines an already correct decision policy rather than learning it from scratch. The KL divergence against the frozen SFT policy separates the two configurations more clearly: with video input it stays lower through the first 60% of training, peaking near 0.010 against 0.015 without video input, after which both decay below 0.006. The judge with video input reaches comparable reward while drifting less from its SFT initialization. The reward standard deviation decreases in both configurations as the sampled

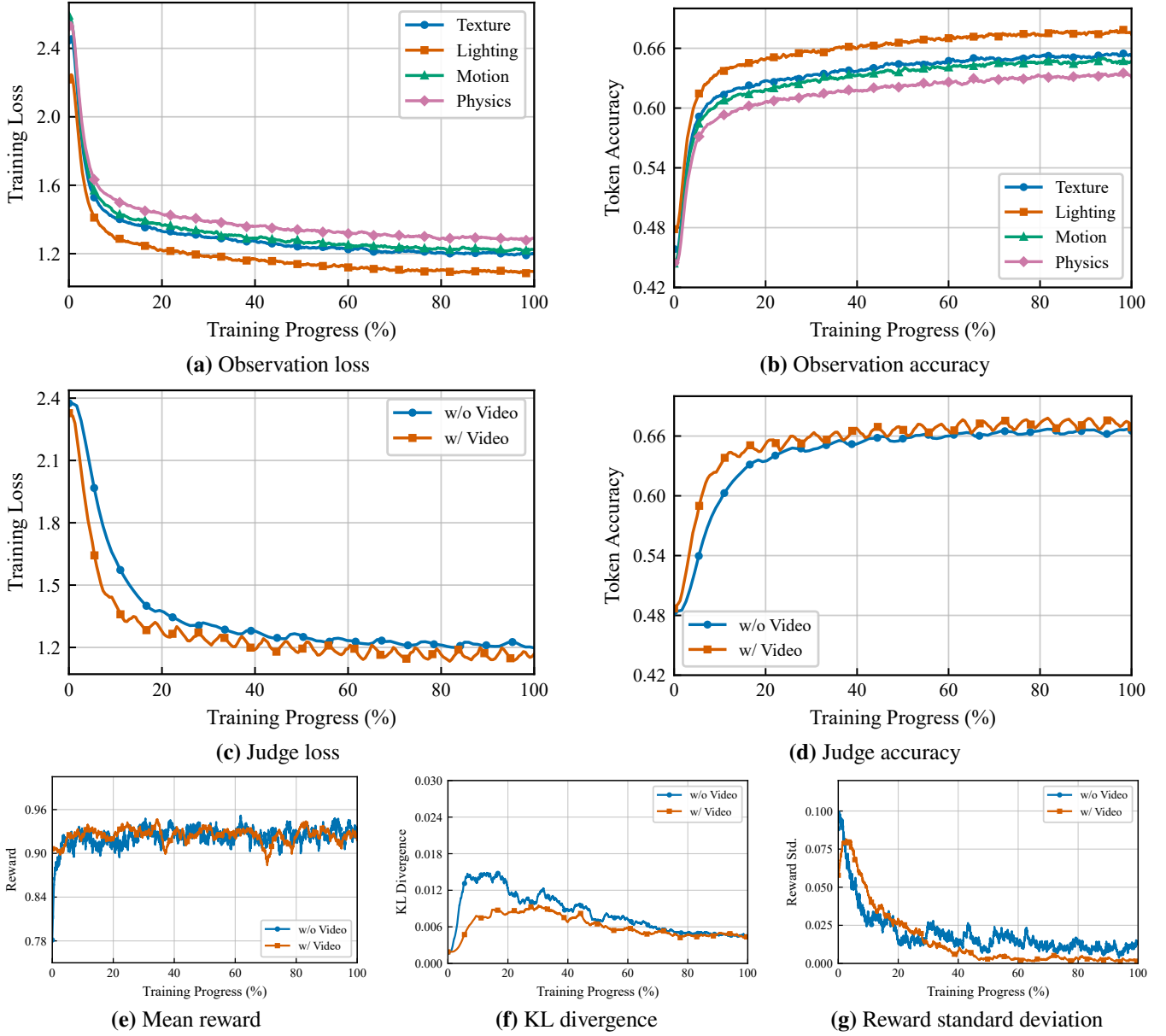


Figure 4: Training dynamics of the framework. (a–d) SFT loss and token accuracy for the four observation agents and for the judge agent with and without video input. (e–g) GRPO mean reward, KL divergence, and reward standard deviation for the judge agent in the same two settings.

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